

Chapter 4 — Convection-dominated problems

Stabilised finite element methods

MTH3340 · Numerical Methods for PDEs

1 Where we are

$$\mu |u - u_h|_{H^1}^2 \leq C(1 + \text{Pe}_h^2) \mu h^{2(r-1)} |u|_{H^r}^2, \quad \text{Pe}_h \doteq \frac{\|\beta\|_{L^\infty} h}{\mu}.$$

- coercivity from μ alone; convection is skew-symmetric;
- accurate only if $\text{Pe}_h \lesssim 1$, i.e. $h \lesssim \mu / \|\beta\|$. Unaffordable.

Read the estimate as an instruction. The constant is fine when

$$\mu \sim \|\beta\|_{L^\infty} h.$$

We cannot change h . Let us change μ . Three doses:

artificial diffusion $\longrightarrow$ streamline diffusion $\longrightarrow$ SUPG

None is a Galerkin method. Each adds a term, and each term has a price.

2 Artificial diffusion

Von Neumann & Richtmyer, Los Alamos, 1950. **Solve with a larger viscosity:**

$$\bar{\mu} \doteq \mu + \mu_{\text{AD}}, \quad \mu_{\text{AD}} \doteq \|\boldsymbol{\beta}\|_{L^\infty(\Omega)} h.$$

Definition (Artificial diffusion method)

Find $u_h \in V_h$ such that, for all $v_h \in V_h$,

$$a_h(u_h, v_h) \doteq \int_{\Omega} \bar{\mu} \nabla u_h \cdot \nabla v_h + \int_{\Omega} v_h (\sigma u_h + (\boldsymbol{\beta} \cdot \nabla) u_h) = \int_{\Omega} f v_h.$$

effective Péclet: $\frac{\|\boldsymbol{\beta}\| h}{\bar{\mu}} = \frac{\|\boldsymbol{\beta}\| h}{\mu + \|\boldsymbol{\beta}\| h} < 1$ for every h .

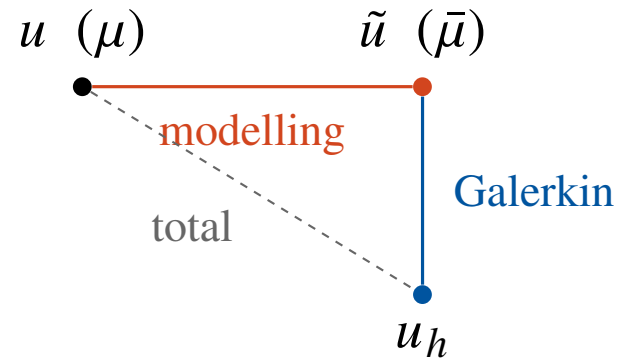
- **stability**: as last week, with $\bar{\mu}$ in place of μ ;
- **but we are solving a different equation.**

Three functions, two errors

$\tilde{u}$: the *continuous* solution with viscosity $\bar{\mu}$,

$$a(\bar{\mu}; \tilde{u}, v) = \ell(v) \quad \forall v \in H_0^1(\Omega).$$

Then u_h is the **plain Galerkin approximation** of $\tilde{u}$, and last week's theorem with $\bar{\mu}$ has a harmless constant: $\|\beta\|h/\bar{\mu} < 1$.



$$u - u_h = \underbrace{(u - \tilde{u})}_{\text{modelling error}} + \underbrace{(\tilde{u} - u_h)}_{\text{Galerkin error: fine}}.$$

Q: How far is $\tilde{u}$ from u ? Redo the error analysis and watch where μ_{AD} shows up.

The consistency error

Exact: $a(\mu; u, v_h) = \ell(v_h)$. Discrete: $a_h(u_h, v_h) = \ell(v_h)$. Subtract:

$$a_h(u - u_h, v_h) = \mu_{\text{AD}} \int_{\Omega} \nabla u \cdot \nabla v_h \quad \forall v_h \in V_h.$$

Galerkin orthogonality is lost. The right-hand side is the consistency error.

Split $u - u_h = e_{Ih} + e_h$, take $v_h = e_h$:

$$a_h(e_h, e_h) = \underbrace{-a_h(e_{Ih}, e_h)}_{\text{as last week, with } \bar{\mu}} + \underbrace{\mu_{\text{AD}} \int_{\Omega} \nabla u \cdot \nabla e_h}_{\text{new}}$$

$$\mu_{\text{AD}} \int_{\Omega} \nabla u \cdot \nabla e_h \leq 2 \mu_{\text{AD}} |u|_{H^1(\Omega)}^2 + \frac{\mu_{\text{AD}}}{8} |e_h|_{H^1(\Omega)}^2 \quad (ab \leq 2a^2 + \frac{1}{8}b^2).$$

The last piece is absorbed by $\bar{\mu} \geq \mu_{\text{AD}}$. The red one stays.

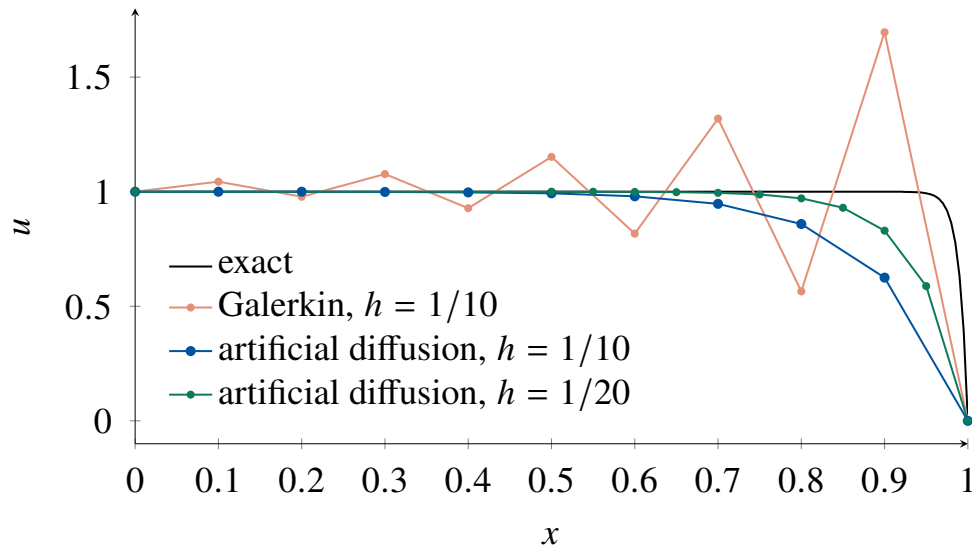
A first-order method, whatever p

$$\bar{\mu} |u - u_h|_{H^1(\Omega)}^2 \lesssim \underbrace{\bar{\mu} h^{2(r-1)} |u|_{H^r(\Omega)}^2}_{\text{interpolation: improves with } r} + \underbrace{\|\beta\|_{L^\infty} h |u|_{H^1(\Omega)}^2}_{\text{consistency: blind to } r}.$$

- **Robust in μ :** the $1/\mu$ constant is gone.
- **Never better than first order in h ,** however high p . Fine for $\mathcal{P}_1$; a waste for $\mathcal{P}_2$ and above.
- **Physically:** viscosity $\|\beta\|h$ smears every layer over a few cells, *in all directions*, also across the flow.

Seeing it in 1D

$$-\mu u'' + u' = 0 \text{ in } (0, 1), \quad u(0) = 1, \quad u(1) = 0, \quad \mu = 10^{-2}, \quad \mathcal{P}_1.$$

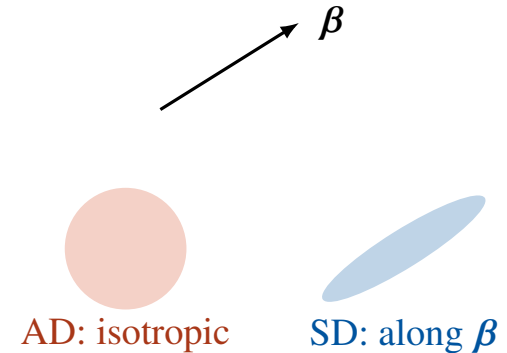


- oscillations **gone**;
- layer of true width 10^{-2} now **spread over 3–4 cells**: width $O(h)$;
- halving h halves the smearing: **first order**.

3 Streamline diffusion

The trouble is with derivatives *along* β . So add diffusion **only along the streamlines**:

$$\tau \int_{\Omega} (\beta \cdot \nabla) u_h (\beta \cdot \nabla) v_h, \quad \tau \doteq \frac{h}{\|\beta\|_{L^\infty}}.$$



Why this τ ? Dimensions. On finite element functions $\|\nabla u_h\| \lesssim h^{-1} \|u_h\|$, so $\nabla \sim h^{-1}$:

$$\text{convection} \sim \|\beta\| h^{-1}, \quad \text{new term} \sim \tau \|\beta\|^2 h^{-2} \quad \implies \quad \tau \sim \frac{h}{\|\beta\|}.$$

In 1D there is no crosswind direction: $\tau \beta^2 u'_h v'_h$ with $\tau \beta^2 = |\beta| h = \mu_{\text{AD}}$.
Streamline diffusion = artificial diffusion.

Streamline diffusion: gain and price

Gain. $v_h = u_h$ gives a new square to control:

$$\tau \|(\boldsymbol{\beta} \cdot \nabla) u_h\|_{L^2(\Omega)}^2 \quad \text{— exactly what cost } 1/\mu \text{ last week.}$$

The convective interpolation term now reads

$$\begin{aligned} - \int_{\Omega} e_{Ih} (\boldsymbol{\beta} \cdot \nabla) e_h &\leq \tau^{-1} \|e_{Ih}\|_{L^2}^2 + \frac{\tau}{4} \|(\boldsymbol{\beta} \cdot \nabla) e_h\|^2, \\ \tau^{-1} \|e_{Ih}\|_{L^2}^2 &\lesssim \|\boldsymbol{\beta}\|_{L^\infty} h^{2r-1} |u|_{H^r}^2. \end{aligned}$$

No μ anywhere. Robust as $\mu \rightarrow 0$.

Price. The exact solution does not satisfy the discrete equations either:

$$a_{\text{SD}}(u - u_h, v_h) = \tau \int_{\Omega} (\boldsymbol{\beta} \cdot \nabla) u (\boldsymbol{\beta} \cdot \nabla) v_h \implies \tau \|(\boldsymbol{\beta} \cdot \nabla) u\|^2 \leq \|\boldsymbol{\beta}\| h |u|_{H^1}^2.$$

Again **$O(h)$, independent of r** . Less dissipative, still first order.

4 SUPG: stabilising without a consistency error

Idea: multiply not u_h but the *residual* of the strong equation, which vanishes for the exact solution.

Definition (Streamline-Upwind Petrov–Galerkin (SUPG), $\sigma = 0$)

Add to the Galerkin form, cell by cell,

$$\sum_{K \in \mathcal{T}_h} \tau_K \int_K \left(-\nabla \cdot (\mu \nabla u_h) + (\boldsymbol{\beta} \cdot \nabla) u_h - f \right) (\boldsymbol{\beta} \cdot \nabla) v_h.$$

- **contains streamline diffusion:** $\tau_K \int_K (\boldsymbol{\beta} \cdot \nabla) u_h (\boldsymbol{\beta} \cdot \nabla) v_h$ is in there;
- **consistent:** for $u \in H^2(\Omega)$ the residual is zero, so **Galerkin orthogonality is back**;
- **why cell by cell?** $u_h \in C^0$ only, so Δu_h lives inside cells, not on Ω .

SUPG stability, part 1: the two extra terms

$v_h = u_h$. The SUPG term is a square plus two leftovers:

$$\underbrace{\sum_K \tau_K \|(\boldsymbol{\beta} \cdot \nabla) u_h\|_{L^2(K)}^2}_{\text{gain}} - \sum_K \tau_K \int_K \left(\underbrace{\nabla \cdot (\mu \nabla u_h)}_{\text{Laplacian}} + \underbrace{f}_{\text{forcing}} \right) (\boldsymbol{\beta} \cdot \nabla) u_h.$$

Forcing: harmless, by Young,

$$\sum_K \tau_K \int_K f (\boldsymbol{\beta} \cdot \nabla) u_h \leq \sum_K 4\tau_K \|f\|_{L^2(K)}^2 + \sum_K \frac{\tau_K}{4} \|(\boldsymbol{\beta} \cdot \nabla) u_h\|_{L^2(K)}^2.$$

Laplacian:

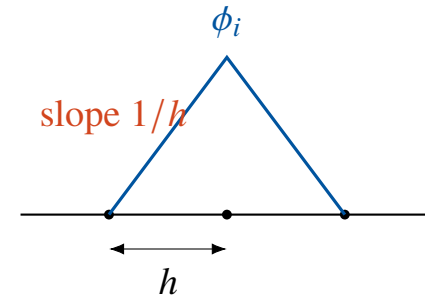
- $\mathcal{P}_1$ on simplices, Q_1 on parallelograms: $\Delta u_h = 0$, nothing to do;
- otherwise compare $\|\Delta u_h\|_K$ with $\|\nabla u_h\|_K$ — an **inverse inequality**.

Inverse inequalities

Hat function in 1D: slope $\pm 1/h$ on a support of length $2h$,

$$\|\phi_i'\|_{L^2}^2 \sim \frac{1}{h^2} \cdot h = h^{-1}, \quad \|\phi_i\|_{L^2}^2 \sim h,$$

$$\|\phi_i'\|_{L^2}^2 \sim h^{-2} \|\phi_i\|_{L^2}^2.$$



Lemma (Discrete inverse inequality)

For u_h in a finite element space of order p on a shape-regular mesh,

$$\|\Delta u_h\|_{L^2(K)} \leq C_{\text{inv}} h_K^{-1} \|\nabla u_h\|_{L^2(K)} \quad \forall K \in \mathcal{T}_h, \quad C_{\text{inv}} \sim p^2.$$

Discrete is the word: false for general H^1 functions. Finite element functions cannot oscillate below the scale h .

SUPG stability, part 2: choosing τ_K

Inverse inequality, then Young ($\|\cdot\|_K \doteq \|\cdot\|_{L^2(K)}$):

$$\begin{aligned} - \sum_K \tau_K \int_K \mu \Delta u_h (\boldsymbol{\beta} \cdot \nabla) u_h &\leq \sum_K (\tau_K^{1/2} h^{-1} \mu C_{\text{inv}} \|\nabla u_h\|_K) (\tau_K^{1/2} \|(\boldsymbol{\beta} \cdot \nabla) u_h\|_K) \\ &\leq \sum_K \tau_K h^{-2} \mu^2 C_{\text{inv}}^2 \|\nabla u_h\|_K^2 + \sum_K \frac{\tau_K}{4} \|(\boldsymbol{\beta} \cdot \nabla) u_h\|_K^2. \end{aligned}$$

The red factor must be absorbed by $\mu \|\nabla u_h\|^2$:

$$\tau_K < \frac{h^2}{4\mu C_{\text{inv}}^2}.$$

One choice, both regimes:

$$\tau_K \doteq \left(\frac{4\mu C_{\text{inv}}^2}{h^2} + \frac{\|\boldsymbol{\beta}\|_{L^\infty(K)}}{h} \right)^{-1} \approx \begin{cases} h/\|\boldsymbol{\beta}\| & \text{convective regime,} \\ h^2/(4\mu C_{\text{inv}}^2) & \text{diffusive regime.} \end{cases}$$

SUPG: the result

Lemma (Stability of SUPG)

With $\tau_K < h^2 / (4\mu C_{\text{inv}}^2)$,

$$\frac{\mu}{4} |u_h|_{H^1(\Omega)}^2 + \sum_K \frac{\tau_K}{2} \|(\boldsymbol{\beta} \cdot \nabla) u_h\|_{L^2(K)}^2 \leq \frac{C_P^2}{2\mu} \|f\|_{H^{-1}(\Omega)}^2 + \sum_K 4\tau_K \|f\|_{L^2(K)}^2.$$

Error. Orthogonality holds, $a_{\text{SUPG}}(e_h, e_h) = -a_{\text{SUPG}}(e_{Ih}, e_h)$, *no consistency term*; streamline control handles convection as before:

$$\mu |u - u_h|_{H^1}^2 + \sum_K \tau_K \|(\boldsymbol{\beta} \cdot \nabla)(u - u_h)\|_{L^2(K)}^2 \lesssim (\mu h^{2(r-1)} + \|\boldsymbol{\beta}\| h^{2r-1}) |u|_{H^r}^2.$$

No $1/\mu$, and no term independent of r : the estimate improves with the order of the space, uniformly in μ . (Full statement and proof in the lecture notes.)

Stable ✓

Consistent ✓

Optimal ✓

Robust as $\mu \rightarrow 0$ ✓

5 Summary

	robust in μ	consistent	order
Galerkin	no	yes	h^{r-1} , constant $\sim 1/\mu$
Artificial diffusion	yes	no	h , smears in all directions
Streamline diffusion	yes	no	h , smears along β only
SUPG	yes	yes	improves with r , robust

In 1D, $\mathcal{P}_1$, $f = 0$: the three stabilised methods coincide. $\Delta u_h = 0$ in every cell, no crosswind direction, $\tau\beta^2 = |\beta|h$. The differences live in 2D (crosswind smearing) and in higher order or $f \neq 0$ (consistency).

Q: Add a reaction term σu_h to the residual. Which extra condition on τ_K does stability need? (Tutorial.)

Next chapter: time-dependent problems and the method of lines.